

Chapter 2

2.2 Assessing Model Accuracy

The most important thing in statistical learning is choosing the right method for your dataset, which depend on these concepts.

2.2.1 Measuring the Quality of Fit

In order to evaluate the performance of a statistical learning method on a given data set, we need some way to measure how well its predictions actually match the observed data.

In the regression setting, the most commonly-used measure is the mean squared error (MSE), given by

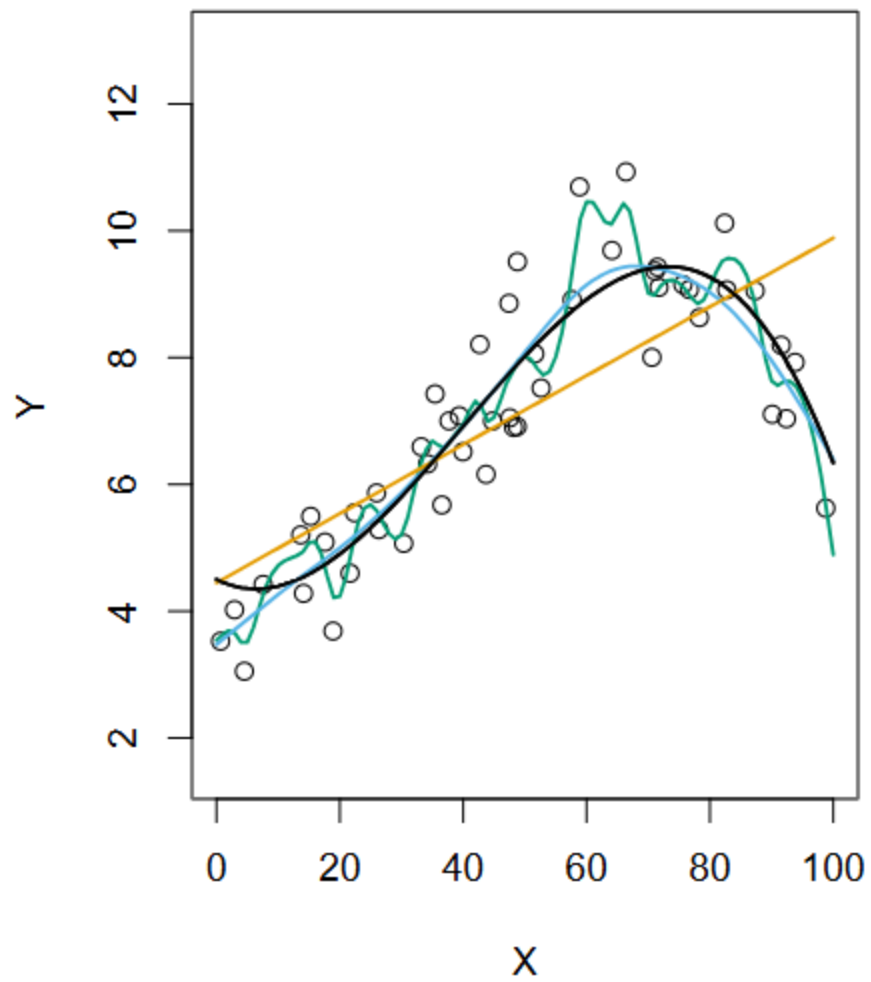
$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}(x_i))^2$$

- $\hat{f}(x_i)$ is the prediction for the i^{th} observation.
- $y_i - \hat{f}(x_i)$ how far is the prediction from the real response.

MSE is computed using Training Data which is referred to as Training MSE.

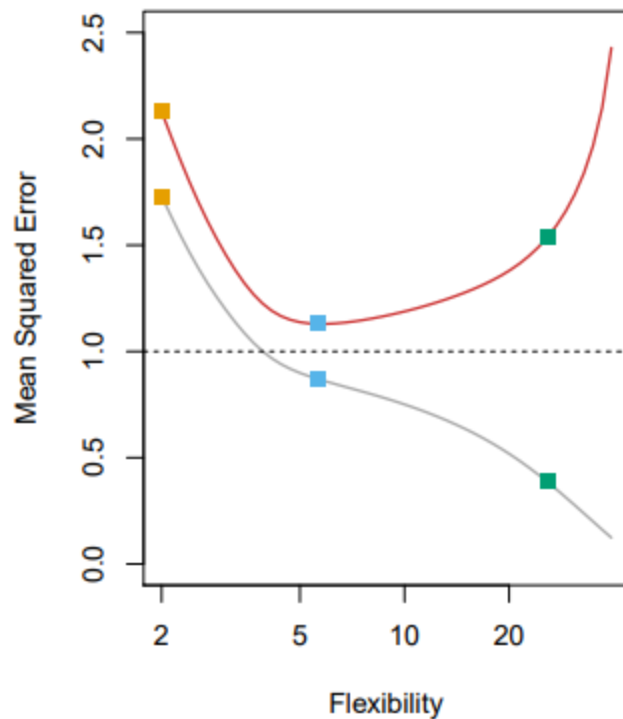
- We do not really care how well the method works training MSE on the Training data but on the **unseen data**.
- y_0 is the prediction for the unseen data (x_0, y_0) .
- We are looking for the method that gives the lowest MSE.
- $\text{Ave}(y_0 - \hat{f}(x_0))^2$

Data from f



- Each circle is an observation.
- Black curve is the real f
- All other curves are an estimate of $f \approx \hat{f}$

Training vs Test MSE



- Training MSE: Grey curve
- Test MSE: Red curve
- Dashed line is the irreducible error $\text{Var}(\epsilon)$

We notice that Test MSE is always bigger than the Training MSE, more flexible model is better at Test MSE results till it reach a point increasing the model flexibility will result on the Test MSE even though it performed best on the Training data, this is also known as Overfitting.

The Bias-Variance Trade-off

The U-shape curve on the Test MSE is the result of Two properties.

1. Variance of $\hat{f}(x_0)$
2. Bias of $\hat{f}(x_0)$
3. Variance of error $\text{Var}(\epsilon)$: irreducible

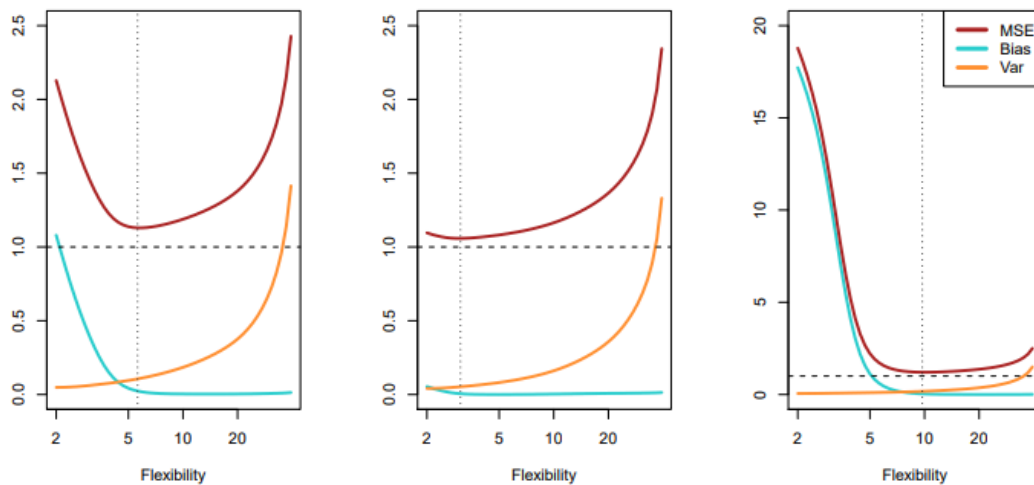
$$\mathbb{E}(y_0 - \hat{f}(x_0))^2 = \text{Var}(\hat{f}(x_0)) + [\text{Bias}(\hat{f}(x_0))]^2 + \text{Var}(\epsilon)$$

- $\mathbb{E}(y_0 - \hat{f}(x_0))^2$ expected Test MSE at x_0
- x_0 unseen observation

To minimize expected Test MSE - we need a model that minimizes Variance and Bias of $\hat{f}$.

- The variance of $\hat{f}$ refers to the amount of change in $\hat{f}$ if we estimated by different data set.

- Different data sets will give different $\hat{f}$
 - High Variance in $\hat{f}$: The smallest change in the data set will result in a very different estimate of $\hat{f}$.
 - $\hat{f}$ shouldn't vary much between different data sets.
 - More flexible methods results in high variance
- The *Bias* of $\hat{f}$ refers to the error introduced by approximating a real life problem which is very complex into simple models.
 - For example Linear regression will assume there is a linear relationship between Y and X, it's unlikely any real life problem is like that.
 - More flexible methods result in lower Bias



- As we increase the flexibility of the model the *Bias* tends to initially decrease.
 - Faster than the *Variance*.
 - As a result of that the Test MSE decline
- At some point increasing the flexibility got no effect on the *Bias*
- But variance will increase significantly (Left and centre Figures)
- In the center figure f what we are estimating is Linear in nature so increasing the flexibility got no effect on the *Bias*.
- In the right figure f is very non-linear that's why we notice a huge drop off in the Bias as the flexibility increases.

The Classification Setting

- When it comes to classification problems $y_1 \dots y_n$ is qualitative
- To quantify the accuracy of estimating $\hat{f}$: Training error rate.

$$\frac{1}{n} \sum_{i=1}^n I(y_i \neq \hat{y}_i)$$

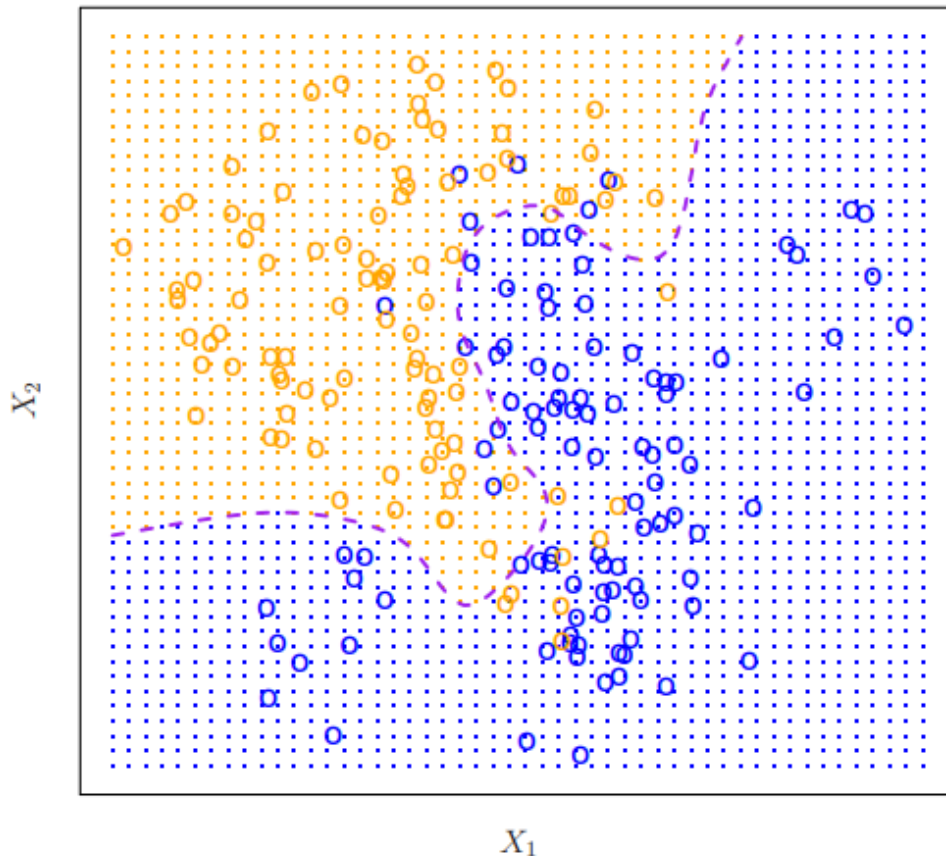
- $\hat{y}_i$ is the predicted category, class label for the i^{th} observation using $\hat{f}$.

- $I(y_i \neq \hat{y}_i)$ indicator variable: 1 if the prediction is wrong $y_i \neq \hat{y}_i$
- And 0 if prediction is correct $y_i = \hat{y}_i$
- It's only for the training data.

We are interested in the *unseen data* error on the observation (x_0, y_0) and it's called test error.

The Bayes Classifier

- Derived from the **Bayes Theorem** bayes classifier.
- Assigns each observation to the most likely class given the predictor value $X = x_0$
- Bayes classifier predicts a class 1 for example if $Pr(Y = 1|X = x_0) > 0.5$



- The orange shaded area the probability $Pr(Y = orange|X) > 50$
- The Blue shaded area the probability $Pr(Y = blue|X) > 50$
- The purple line represents the points where probability is exactly 50%: **Bayes decision boundary**
- The Bayes classifier produces the lowest possible test error rate
- And always pick the class with the highest value

The error rate will be:

$$1 - E(\max_j Pr(Y = j|X))$$

- The Bayes classifier is used in K-Nearest Neighbours to predict the most likely class k for a given observation.